



ELSEVIER

Information Processing Letters 75 (2000) 113–117

Information
Processing
Letters

www.elsevier.com/locate/ipl

Computational complexity of similarity retrieval in a pictorial database

D.J. Guan^{a,*}, Chun-Yen Chou^b, Chiou-Wei Chen^b

^a Department of Computer Science, National Sun Yat-Sen University, Kaohsiung 80424, Taiwan

^b Department of Applied Mathematics, National Sun Yat-Sen University, Kaohsiung 80424, Taiwan

Received 12 January 2000; received in revised form 5 April 2000

Communicated by K. Iwama

Abstract

Three types of similarity are commonly used in similarity retrieval of a pictorial database. Only one of them is isometric, hence transitive. We present polynomial time algorithms to find the maximum similar subpicture of two given pictures when the similarity is isometric. For nonisometric types, we show that finding a maximum similar subpicture in two given pictures is NP-complete, even when all symbols are distinct, by reducing the satisfiability problem to it. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Information retrieval; Pictorial database; Similarity retrieval

1. Introduction

Storing and retrieving of pictures in a database can be quite different than in conventional databases that handle only texts. To be able to retrieve pictures by their contents, descriptions of the pictures are often stored together with the pictures. Tanimoto introduced iconic indexing to describe the contents of a picture [5]. In short, symbols and their coordinates are used to describe a picture. This allows the use of a spatial relationship of symbols for the retrieval of pictures. For example, we may specify a picture which has a tree in front of a house.

The picture itself and the symbolic description of the picture are two different entities stored in a

pictorial database. In this paper, a picture is defined as the symbolic description of the picture, not the picture itself.

Spatial relationship is a fuzzy concept depending on human interpretation. Thus, the specifications of pictures requested by a user may not be exact. Similarity retrieval of pictures, which is one of the distinguishing functions different for a conventional database, must be provided in a pictorial database. There may be many criteria to measure the “similarity” between two given pictures. In this paper, we consider the case of finding subpictures of maximum size which are “similar”. We will define this criterion formally in the following paragraphs.

In general, there are eight directions which are commonly used in two-dimensional pictures, namely left, right, up, down, upper left, upper right, lower left, and lower right. According to these eight directions, given two pictures P_1 and P_2 , Chang et al. defined

* Corresponding author.

E-mail addresses: guan@cse.nsysu.edu.tw (D.J. Guan),
chou@math.nsysu.edu.tw (C.-Y. Chou).

three types of similarity relation for any pair of objects in P_1 and P_2 [1]. The three types of similarity are equivalent to the following definitions by using the coordinates of the symbols.

Let A be the symbol assigned to an icon whose coordinates are (x, y) in a picture P . We call the symbol A with coordinates (x, y) an *object* of P , and it is denoted by $[A, (x, y)]$.

Definition 1. Let $[A, (x_1, y_1)]$ and $[B, (x_2, y_2)]$ be two objects in P_1 , and $[A, (a_1, b_1)]$ and $[B, (a_2, b_2)]$ two objects in P_2 . Let $\Delta_x = x_2 - x_1$, $\Delta_y = y_2 - y_1$, $\Delta_a = a_2 - a_1$, and $\Delta_b = b_2 - b_1$. We say that objects $[A, (x_1, y_1)]$ and $[B, (x_2, y_2)]$ in P_1 and objects $[A, (a_1, b_1)]$ and $[B, (a_2, b_2)]$ in P_2 have a relation of type- i , $i = 0, 1, 2$, according to the corresponding criterion.

type-0: $\Delta_x \Delta_a \geq 0$ and $\Delta_y \Delta_b \geq 0$.

type-1: “ $\Delta_x \Delta_a > 0$ or $\Delta_x = \Delta_a = 0$ ” and “ $\Delta_y \Delta_b > 0$ or $\Delta_y = \Delta_b = 0$ ”.

type-2: $\Delta_x = \Delta_a$ and $\Delta_y = \Delta_b$.

Definition 2. Let S be a subset of objects in picture P_1 and T be a subset of objects in P_2 . We say that S and T are similar subpictures of type- i , $i = 0, 1, 2$, respectively, if there is a bijective function f from S to T such that:

- (1) For every object $[A, (x, y)]$ in S , the first component of $[A, (x, y)]$ and the first component of $f([A, (x, y)])$ are the same. That is, if we consider a picture to be a set of icons [5], they represent the same type of icons.
- (2) For any pair of objects $[A, (x_1, y_1)]$ and $[B, (x_2, y_2)]$ in S , $[A, (x_1, y_1)]$ and $[B, (x_2, y_2)]$ in P_1 and $f([A, (x_1, y_1)])$ and $f([B, (x_2, y_2)])$ in P_2 have the spatial relation of type- i , $i = 0, 1, 2$, respectively.

The problem of finding maximum similar subpictures in P_1 and P_2 is to find a subset of objects S in P_1 and a subset of objects T in P_2 such that S and T are similar subpictures of type- i , $i = 0, 1, 2$, with maximum cardinality.

Intuitively, in finding maximum similar subpictures, the case when the symbols are all distinct is easier than the case when the symbols are not all distinct.

Given two pictures P_1 and P_2 . When symbols are all distinct in each picture, Chang et al. reduce the problem of finding maximum similar subpictures of type- i , $i = 0, 1, 2$, to the clique problem of a graph constructed from P_1 and P_2 . A graph $G_i(P_1, P_2)$, $i = 0, 1, 2$, is constructed as follows. Let the vertex set V be the set of symbols appearing in both P_1 and P_2 . There is an edge between A and B if and only if there are two objects $[A, (x_1, y_1)]$, $[B, (x_2, y_2)]$ in P_1 , and two objects $[A, (a_1, b_1)]$, $[B, (a_2, b_2)]$ in P_2 , such that they have the spatial relation of type- i , $i = 0, 1, 2$, respectively. Therefore we have an undirected graph $G_i(P_1, P_2)$, and a maximum complete subgraph of $G_i(P_1, P_2)$ corresponds to a maximum similar subpicture of type- i , $i = 0, 1, 2$.

Note that similarity of type-2 is isometric, that is, the difference in both coordinates are preserved, hence transitive. For maximum similar subpictures of type-2, we present an $O(n^2)$ algorithm when symbols are all distinct and an $O(n^3)$ algorithm when symbols are not all distinct.

A related problem is called the *picture pattern matching problem*. Given a picture P and a pattern Q , determine whether the pattern Q appears in the picture P or not. Tucci et al. [6] showed that type-0 and type-1 picture matching problems are NP-complete when the symbols are not all distinct.

Note that when all the symbols are distinct, we can construct the graph $G_i(P, Q)$ and check if the induced subgraph $G_i(P, Q)[U]$ is a complete graph or not, where U is the subset of vertices corresponding to the pattern Q . This shows that the picture pattern matching problem can be solved in polynomial time when all the symbols are distinct.

We show in this paper that picture similarity problems of type-0 and type-1 are NP-complete even when the symbols are all distinct. Our method is to reduce the satisfiability problem to them.

2. Maximum similar subpictures of type-2

Assume that we are given two pictures P_1 and P_2 each containing a set of n distinct objects. Recall that, to find a maximum similar subpicture of type-2, we can construct the corresponding undirected graph $G_2(P_1, P_2)$. We show that $G_2(P_1, P_2)$ is a union of complete subgraphs. Therefore, there is an efficient

algorithm for finding maximum similar subpictures of type-2.

Note that similarity of type-2 is transitive. Let $[A, (x_1, y_1)]$, $[B, (x_2, y_2)]$, and $[C, (x_3, y_3)]$ be three distinct objects in P_1 , and $[A, (a_1, b_1)]$, $[B, (a_2, b_2)]$, and $[C, (a_3, b_3)]$ be three distinct objects in P_2 . Since all symbols are distinct, we may use the short notation “ A and B have a relation of type-2” to mean that “ $[A, (x_1, y_1)]$ and $[B, (x_2, y_2)]$ in P_1 and $[A, (a_1, b_1)]$ and $[B, (a_2, b_2)]$ in P_2 have a relation of type-2”. Assume that

- (1) A and B have a relation of type-2, and
- (2) B and C have a relation of type-2.

We can conclude that A and C must have a relation of type-2. In the graph $G_2(P_1, P_2)$, if there is an edge between A and B and an edge between B and C , then there must be an edge between A and C . Therefore, the graph $G_2(P_1, P_2)$ is a union of complete subgraphs. Hence, by checking the adjacency lists, it only takes $O(n)$ time to identify each maximal complete subgraph. Since it only takes $O(n^2)$ time to construct the graph $G_2(P_1, P_2)$, we have the following theorem.

Theorem 1. *The problem of finding maximum similar subpictures of type-2 can be solved in $O(n^2)$ time when all symbols are distinct.*

When symbols are not all distinct, we have the following algorithm, taking $O(n^3)$ time, to solve the problem of maximum similar subpictures of type-2.

The algorithm first lists all objects in P_1 and all objects in P_2 , respectively, in the order of the sum of their horizontal and vertical coordinates and, when there is a tie, in the order of horizontal coordinates. The algorithm then finds a maximum subset of objects S in P_1 and T in P_2 , such that S and T are similar subpictures. Suppose that $[A, (x_1, y_1)]$ in S is mapped to $[A, (a_1, b_1)]$ in T under f , that is, $f([A, (x_1, y_1)]) = [A, (a_1, b_1)]$. Then any object $[B, (x_2, y_2)]$ in P_1 and $[B, (a_2, b_2)]$ in P_2 with $x_2 - x_1 = a_2 - a_1$ and $y_2 - y_1 = b_2 - b_1$ must also be in S and T , respectively, and $f([B, (x_2, y_2)]) = [B, (a_2, b_2)]$. We shall call the two objects $[B, (a_2, b_2)]$ in P_1 and $[B, (x_2, y_2)]$ in P_2 a *match* with respect to the pair of objects $[A, (x_1, y_1)]$ in S and $[A, (a_1, b_1)]$ in T .

The algorithm computes S and T by superimposition. For each $[s_i, (x_i, y_i)]$ in P_1 , let Q_{s_i} be the subset

of objects in P_2 whose first component is s_i . Then, for each $[s_i, (a_j, b_j)]$ in Q_{s_i} , the algorithm counts the number of matches by superimposing P_1 and P_2 . The superimposing is done by identifying the coordinates (x_i, y_i) of P_1 and (a_j, b_j) of P_2 . That is, subtract $\Delta_x = a_j - x_i$ from all horizontal coordinates of objects in P_2 , and subtract $\Delta_y = b_j - y_i$ from all vertical coordinates of objects in P_2 . Then the algorithm counts the number of matches in the two modified lists. Since the original objects in each picture are ordered, the counting of the number of matches can be done in linear time by scanning the two lists in a way similar to merge sort.

It is easy to see that, for each object $[s_i, (x_i, y_i)]$ in P_1 and each $[s_i, (a_j, b_j)]$ in Q_{s_i} , it takes $O(n)$ time to count the matches. Since the number of objects in P_1 is at most n , the number of objects in Q_{s_i} is at most n for each s_i , the counting of matches is performed at most n^2 times. Therefore the above algorithm is of $O(n^3)$ time.

Theorem 2. *The problem of finding maximum similar subpictures of type-2 can be solved in $O(n^3)$ time when symbols are not all distinct.*

3. Maximum similar subpicture problem of type-0 or type-1

In this section, we show that finding the maximum similar subpicture of type-0 or type-1 is NP-complete. We reduce the satisfiability problem to them.

Let the set of Boolean variables be $\{x_1, x_2, \dots, x_k\}$. Given a Boolean formula Φ in conjunctive normal form, say, $\Phi = c_1 \wedge c_2 \wedge \dots \wedge c_n$, where for each clause c_r , $r = 1, 2, \dots, n$, $c_r = (l'_1 \vee l'_2 \vee l'_3)$ for some distinct literals l'_1, l'_2, l'_3 . We construct two pictures P_1 and P_2 from Φ as follows.

P_1 and P_2 are both of size $3n$ in width and $2k$ in height. Each picture will contain $3n$ objects, and they are labeled as l'_r, l''_r, l'''_r , $r = 1, 2, \dots, n$. The coordinates of each object with symbol l'_i in each picture are assigned as follows.

- In picture P_1 : $[l'_i, (g_1(i, r), h_1(i, r))]$.
- In picture P_2 : $[l'_i, (g_2(i, r), h_2(i, r))]$.

The functions g_1, g_2, h_1 , and h_2 are defined as follows.

$$g_1(i, r) = \begin{cases} 3r - 2 & \text{if } i = 1, \\ 3r - 1 & \text{if } i = 2, \\ 3r & \text{if } i = 3, \end{cases}$$

$$g_2(i, r) = \begin{cases} 3r & \text{if } i = 1, \\ 3r - 1 & \text{if } i = 2, \\ 3r - 2 & \text{if } i = 3, \end{cases}$$

$$h_1(i, r) = \begin{cases} 2j & \text{if } l_i^r \text{ is the Boolean variable } \bar{x}_j, \\ 2j - 1 & \text{if } l_i^r \text{ is the Boolean variable } x_j, \end{cases}$$

$$h_2(i, r) = \begin{cases} 2j - 1 & \text{if } l_i^r \text{ is the Boolean variable } \bar{x}_j, \\ 2j & \text{if } l_i^r \text{ is the Boolean variable } x_j. \end{cases}$$

The symbols are l_i^r 's, hence all distinct. Note that it takes polynomial time to construct P_1 and P_2 .

Example. If $\Phi = (x_1 \vee \bar{x}_3 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2 \vee x_3)$ with the specified order of clauses and literals, then the corresponding P_1 is

	l_2^1				
					l_3^2
		l_3^1			
				l_2^2	
			l_1^2		
l_1^1					

The corresponding P_2 is

			l_3^2		
	l_2^1				
				l_2^2	
l_3^1					
		l_1^1			
					l_1^2

Between P_1 and P_2 , the only differences are

- (1) within each clause, the order of columns for the three literals are opposite;

- (2) for two complementary literals, the order of rows are opposite.

Therefore we have the following properties.

Property I. For the same r and $i \neq j$, the two objects $[l_i^r, (g_1(i, r), h_1(i, r))]$, $[l_j^r, (g_1(j, r), h_1(j, r))]$ in P_1 and the two objects $[l_i^r, (g_2(i, r), h_2(i, r))]$, $[l_j^r, (g_2(j, r), h_2(j, r))]$ in P_2 don't have a relationship of type-0 or type-1.

Property II. For $r \neq s$ and any i, j , if the Boolean variables for l_i^r and l_j^s are complementary, then the two objects $[l_i^r, (g_1(i, r), h_1(i, r))]$, $[l_j^s, (g_1(j, s), h_1(j, s))]$ in P_1 and the two objects $[l_i^r, (g_2(i, r), h_2(i, r))]$, $[l_j^s, (g_2(j, s), h_2(j, s))]$ in P_2 don't have a relationship of type-0 or type-1.

Property III. For $r \neq s$ and any i, j , if the Boolean variables for l_i^r and l_j^s are not complementary, then the two objects $[l_i^r, (g_1(i, r), h_1(i, r))]$, $[l_j^s, (g_1(j, s), h_1(j, s))]$ in P_1 and the two objects $[l_i^r, (g_2(i, r), h_2(i, r))]$, $[l_j^s, (g_2(j, s), h_2(j, s))]$ in P_2 have a relationship of type-0 or type-1.

Clearly, by Property I, any subset of objects with a relation of type-0 or type-1 between P_1 and P_2 has at most one literal from each clause. Therefore its size is at most n . For brevity, we use $\max_i \{P_1, P_2\}$, $i = 0, 1$, to denote the maximum size of subsets having a relation of type- i between P_1 and P_2 . Clearly, $\max_i \{P_1, P_2\} \leq n$, $i = 0, 1$.

Lemma 1. Given two pictures P_1 and P_2 .

- (1) Φ is satisfiable if and only if $\max_0 \{P_1, P_2\} = n$.
- (2) Φ is satisfiable if and only if $\max_1 \{P_1, P_2\} = n$.

Proof. If Φ is satisfiable, then there is a satisfying assignment and each clause c_r contains at least one literal l_i^r assigned true. Pick one such literal from each clause, and add the corresponding object $[l_i^r, (g_1(i, r), h_1(i, r))]$ in P_1 to S and the corresponding object $[l_i^r, (g_2(i, r), h_2(i, r))]$ in P_2 to T , we get a set S and T of size n . Note that for any two objects $[l_i^r, (g_1(i, r), h_1(i, r))]$ and $[l_j^s, (g_1(j, s), h_1(j, s))]$ in S , the Boolean variables for l_i^r and l_j^s are not complementary. Therefore the two objects $[l_i^r, (g_1(i, r), h_1(i, r))]$, $[l_j^s, (g_1(j, s), h_1(j, s))]$ in S and the two objects $[l_i^r, (g_2(i, r), h_2(i, r))]$, $[l_j^s, (g_2(j, s), h_2(j, s))]$ in T have a relationship of type-0 or type-1.

in T have a relationship of type-0 by Property III. Hence S and T are similar subpictures of type-0. Therefore $\max_0\{P_1, P_2\} = n$.

If there is a subset S of objects in P_1 with size n , and a subset of objects T in P_2 such that S and T are similar subpictures of type-0, then for each clause c_r , S contains an object $[l_i^r, (ig_1(i, r), h_1(i, r))]$ where l_i^r is a literal in the clause c_r by Property I and the pigeonhole principle. For any two objects $[l_i^r, (ig_1(i, r), h_1(i, r))]$ and $[l_j^s, (ig_1(j, s), h_1(j, s))]$ in S , the Boolean variables for l_i^r and l_j^s are not complementary by Property II. Therefore, by assigning true to each corresponding literal in S , Φ is satisfiable.

Similarly, we can show that Φ is satisfiable if and only if $\max_1\{P_1, P_2\} = n$. $\square$

Therefore, we have the following two lemmas.

Lemma 2.

- (1) *The problem of finding maximum similar subpictures of type-0 is NP-hard.*
- (2) *The problem of finding maximum similar subpictures of type-1 is NP-hard.*

Recall that, when symbols are all distinct, we can reduce the problem of maximum similar subpictures of type-0 (respectively, type-1) to the maximum clique

problem. Therefore, both problems are in NP. We then have the following conclusions.

Theorem 3.

- (1) *The problem of finding maximum similar subpictures of type-0 is NP-complete even when all symbols are distinct.*
- (2) *The problem of finding maximum similar subpictures of type-1 is NP-complete even when all symbols are distinct.*

References

- [1] S.K. Chang, Q.Y. Shi, C.W. Yan, Iconic indexing by 2D strings, IEEE Trans. Pattern Anal. Machine Intelligence 9 (3) (1987) 413–428.
- [2] S.K. Chang, C.W. Yan, D.C. Dimitroff, T. Arndt, An intelligent image database system, IEEE Trans. Software Engrg. 14 (5) (1988) 681–688.
- [3] G. Costagliola, F. Ferrucci, G. Tortora, M. Tucci, Correspondences—Non-redundant 2D strings, IEEE Trans. Knowledge Data Engrg. 7 (2) (1995) 347–350.
- [4] S.Y. Lee, M.K. Shan, W.P. Yang, Similarity retrieval of iconic image database, Pattern Recognition 22 (1988) 675–682.
- [5] S.L. Tanimoto, An iconic symbolic data structuring scheme, in: Pattern Recognition and Artificial Intelligence, Academic Press, New York, 1976, pp. 452–471.
- [6] M. Tucci, G. Costagliola, S.K. Chang, A remark on NP-completeness of picture matching, Inform. Process. Lett. 39 (1991) 241–243.